

CSE260 Assignment-3

The following questions are graded and need to be submitted

Question 1(Graded):

Design a circuit diagram for the following system that takes a 4-bit binary number [A] as inputs and outputs in the following fashion:

If A is odd: The output should be A-4

If A is even: The Output should be A+4

Question 2(Graded) :

Construct the function $F(a,b,c,d) = \sum(0, 1, 6, 10, 12, 14, 15)$:

- Using 3x8 decoders and 2x4 decoders (you need to use both in the same circuit)
- Using 4x1 and 2x1 mux (you need to use both in the same circuit)

Your circuit must be efficient, which means you need to use the least amount of components.

Question 3(Graded):

Design a 13 person voter counting system using full and parallel adders. You must use encoder and decoder to build full adders, meaning you can't use full adders directly.

Question 4(Graded):

Spider-Man is keeping an eye on a big city march. He has four tiny sensors set up around the area. Each sensor can send a "1" if it spots something important:

- D3 – Rooftop scout: sees leaders forming at the front
- D2 – Street camera: sees group movement in the middle
- D1 – Alley microphones: hear shouts or calls for help from side streets
- D0 – Wearable devices: distress from individual civilians

He needs a device that **picks the most important signal** at any moment, and outputs it as a **2-bit code**. This is done using a 4×2 priority encoder — but with special rules.

The Rules for Priority:

- If there is No "1" signal → output should be don't care.
- Only two or more consecutive "1"s in a row → Treat as a "group event" and The priority goes to the leftmost 1 in that group — the one with the bigger bit position (closer to D3)
- Exactly two "1"s but not consecutive → The priority goes to the rightmost 1 in that group — the one with the bigger bit position (closer to D0)
- Anything else - pick D0

Now, design the circuit diagram of the priority encoder and use K-Map to simplify the function.

The questions below are ungraded and are not required to submit

1. Design a 13 people voting system using necessary parallel and full adder(s). Your circuit diagram should be efficient, i.e. use the least amount of components.
2. Design a **BCD to excess 4** system using necessary parallel adder(s).
3. Construct the function $F(a,b,c,d) = \Sigma(0, 1, 2, 5, 7, 9, 10, 15)$:
 - a. Using only 3x8 decoders
 - b. Using 3x8 decoder(s) and 2x4 decoder(s) (you need to use both in the same circuit)
 - c. Using only 2x4 decoders

You may use basic gates in every case. Your circuit must be efficient, which means you need to use the least amount of components.

4. Construct the function $F(a,b,c,d,e) = \Sigma(0,1, 5, 8, 9, 13, 16, 19, 24, 25, 31)$:
 - a. Using 16x1 and 2x1 mux(s) (you need to use both in the same circuit)
 - b. Using only 8x1 muxes
 - c. Using 8x1 and 4x1 mux(s) (you need to use both in the same circuit)

Your circuit must be efficient, which means you need to use the least amount of components.

5. Construct the following circuits using only 4x1 mux(s):
 - a. $ab' + c$
 - b. $(x' + y)(y + z')$
6. You have to construct the truth table and internal circuit diagram for a **4x2 priority encoder**. The priority is given in the following manner:
 - a. If the number of 1 in a binary input is even, then priority will be given to the MSB line.
 - b. If the number of 1 in a binary input is odd, then priority will be given to the LSB line.

7. Build a circuit that implements the 1's complement number system (3 bits) using encoder(s) and decoder(s).
8. Implement the following boolean function using **a)** single 16:1 mux **b)** single 8:1 mux
 $F(A,B,C,D) = \Sigma(0,1,2,7,8,10,11,13, 15)$. Use external gates if required
9. Implement the following boolean function using **a)** 4x16 decoder(s) only **b)** 2x4 decoder(s) only
 $F(A,B,C,D,E) = \Sigma(0,1,2,7,8,10,11,13, 15,18,21,24,25)$. Use external gates if required.